MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2026 MID SEMESTER ONE EXAMINATIONS

MATH 1203: DISCRETE MATHEMATICS (ODEL)

DATE: 25$^{\text{th}}$ March, 2026 **TIME**: 1:00 - 3:00

EXAMINER: M. Kanyika

Instructions

1. This paper contains 3 printed pages, including the cover page.Please check.
2. Attempt **all 4** questions.
3. Please write clearly, legibly and show all your working.
4. Begin each question on a fresh page.
5. Write down your **Registration number** at the top of each page of the answer sheet.
6. Use of non-programmable calculators is allowed.

Question	1	2	3	4	Total
Marks	15	17	14	16	62

DO NOT TURN OVER THE QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

Question 1

(a) State the Pigeon Hole Principle (2 **marks**)

(b) What is the minimum number of students, each of whom comes from one of the 28 districts in Malawi, who must be enrolled in a university to guarantee that there are at least 12 who come from the same district? (3 **marks**)

(c) Search for 23 in the list 1 2 3 5 6 7 8 10 12 13 15 16 19 21 23 24 using binary search (5 **marks**)

(d) Find the minimum number of coins needed to make up a target amount of 93 cents (5 **marks**)

Question 2

(a) How many license plates can be made using either two letters followed by three digits or two digits followed by four letters? (4 **marks**)

(b) How many bit of strings are there of length eight, either start with bits **000** or end with bits **11**? (5 **marks**)

(c) A CSS & BIT online portal requires students to create access codes that are between 4 and 6 characters long, where each character can be either an uppercase letter (A–Z) or a digit (0–9). Each access code must contain at least one letter and at least one digit, so codes consisting entirely of letters or entirely of digits are not allowed. Repeating characters are permitted. Determine the total number of valid access codes that satisfy these conditions, taking into account all possible lengths, and show all your working. (8 **marks**)

Question 3

(a) Find the number of bit strings of length 4 with no consecutive 1s. (4 **marks**)

(b) The Department of Mathematics at the Malawi University of Science and Technology prepares examinations for three different student groups: ODEL, Weekend, Generic.For each group, the department may set one of three types of examination papers: EOS (End of Semester), Deferred, Supplementary.Each paper consists of a fixed number of questions. The Weekend and Generic groups each receive papers containing 8 questions, while the ODEL group receives papers containing 5 questions. Assume that: Each combination of group and paper type represents a distinct examination.Each question on a paper is considered distinct and contributes to the structure of that examination.

i. Draw a tree diagram to represent the possible examination structures. (6 **marks**)

ii. Determine the total number of distinct examination structures that can be produced. (4 **marks**)

Question 4

(a) Use iteration to solve the recurrence relation $a_n = a_{n-1} + 2$ where $a_0 = 1$, for $n \geq 1$. (3 **marks**)

(b) Show that the sequence $a_n = (-4)^n$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$. (5 **marks**)

(c) Find all solutions of the recurrence relation $y = 5y_{n-1} - n, \quad n \geq 2$ with initial condition $y_1 = 0$ (8 **marks**)

End of question paper